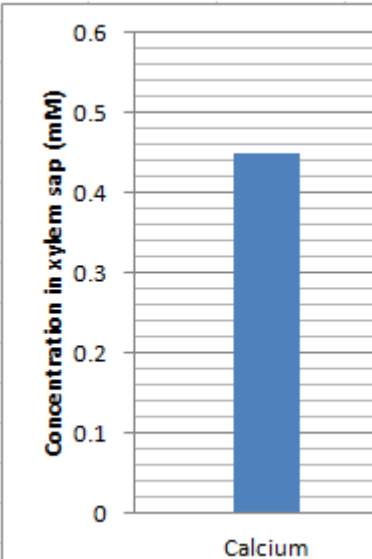


Question Number	Answer	Mark
1 (a)(i)	The only correct answer is D Z <i>A is incorrect because it is not a xylem vessel</i> <i>B is incorrect because it is not a xylem vessel</i> <i>C is incorrect because it is not a xylem vessel</i>	(1)

Question Number	Answer	Mark
1(a)(ii)	The only correct answer is B V <i>A is incorrect because it is not a sclerenchyma fibre</i> <i>C is incorrect because it is not a sclerenchyma fibre</i> <i>D is incorrect because it is a xylem vessel</i>	(1)

Question Number	Answer	Additional guidance	Mark				
1(b)(i)	An answer showing:  <table border="1"><thead><tr><th>Element</th><th>Concentration in xylem sap (mM)</th></tr></thead><tbody><tr><td>Calcium</td><td>0.45</td></tr></tbody></table>	Element	Concentration in xylem sap (mM)	Calcium	0.45	Plotted top of the bar must not touch a printed grid line above or below 0.45 IGNORE the width of the bar	(1)
Element	Concentration in xylem sap (mM)						
Calcium	0.45						

Question Number	Answer	Additional guidance	Mark
1(b)(ii)	An explanation that includes the following points: - calcium ions to make calcium pectate to {hold cells together / increase cell wall stability / form middle lamella} (1) - magnesium ions to make chlorophyll so {photosynthesis can occur / light energy can be absorbed / glucose can be made} (1) - nitrates are needed to make {amino acids / proteins / polypeptides / DNA / RNA / nucleic acid} (1)	Accept calcium ions to make calcium pectate to increase cell wall strength Ignore make cell walls Ignore chloroplasts Accept 'sugars'	(3)